Value Theorems of Calculus — Revision Summary

IVT, Rolle, Lagrange, Cauchy, and Darboux at a glance

1 Intermediate Value Theorem (IVT)

Theorem (IVT). *If $f : [a, b] \to \mathbb{R}$ is continuous and y lies between $f(a)$ and $f(b)$, then $f(c) = y$ for some $c \in [a, b]$.*

Intuition. A continuous curve cannot skip values — a statement about *connectedness*. The most-used special case: opposite signs at the endpoints force a root inside.

When to reach for it. Anywhere you must prove *something exists* — a root, a fixed point, a meeting — with no differentiability needed. The classic move: build an auxiliary $h(x) = f(x) - g(x)$ whose zeros are exactly what you want.

Key applications.

- **Root location:** Sign-change scans locate roots of polynomials and transcendental equations without solving them.
- **Fixed-point theorem:** Any continuous $f : [0, 1] \to [0, 1]$ has a fixed point. Apply IVT to $g(x) = f(x) - x$. (One-dimensional Brouwer.)
- **Antipodal points / meeting problems:** Two hikers walking toward each other must meet; antipodal points on the equator share temperature; two monks ascending and descending on different days cross paths. All come from $h(x) = f(x) - f(x + \text{shift})$ being odd in some sense.
- **Odd-degree polynomials always have a real root:** Their values at $\pm\infty$ have opposite signs.

1.1 The intermediate-value property (IVP) and discontinuity

The converse of IVT *fails*: a function can have IVP without being continuous. To see why, classify discontinuities at x_0 (let $L^{\pm}$ denote the one-sided limits):

- **Removable:** $L^- = L^+ = L$, but $L \neq f(x_0)$. A pluggable hole.
- **Jump:** L^-, L^+ both exist but differ. A clean leap.
- **Essential:** at least one one-sided limit fails to exist (typically wild oscillation).

The first two are called *simple* (or first-kind) discontinuities; the third is *second-kind*.

Theorem (IVP rules out simple discontinuities). *If f has IVP on an interval, its only possible discontinuities are essential.*

Intuition. A jump or removable discontinuity creates a value the function locally misses; only oscillation lets the function hit every nearby value despite being discontinuous.

Canonical example. $f(x) = \sin(1/x)$ for $x \neq 0$, $f(0) = 0$. Essentially discontinuous at 0, yet possesses IVP everywhere because it oscillates through every value in $[-1, 1]$ in every neighbourhood of 0.

1.2 Darboux's Theorem — derivatives have IVP

Theorem (Darboux). *If f is differentiable on $[a, b]$, then f' has the intermediate-value property: for any y between $f'(a)$ and $f'(b)$, some $c \in (a, b)$ has $f'(c) = y$.*

Why it matters. Derivatives can be discontinuous (e.g., the derivative of $x^2 \sin(1/x)$, extended by 0 at 0, oscillates wildly near 0). But Darboux says: even discontinuous derivatives must satisfy IVP. Combined with the previous theorem, this means *a derivative's only allowed discontinuities are essential.* No jumps, no removable holes.

Diagnostic use. "Being a derivative" is a restrictive property. To show a function g is *not* the derivative of anything, exhibit a value it skips between two of its values. The Heaviside step function, for instance, isn't anyone's derivative.

2 Rolle's Theorem

Theorem (Rolle). *If f is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.*

Intuition. A hiker who starts and ends at the same altitude must, somewhere, momentarily be moving level. Equivalently: an interior extremum exists, and at it $f' = 0$ (Fermat).

When to reach for it. Whenever you can spot — or *engineer* — two equal values of a function, and you want a zero of the derivative. The strongest signals:

- **Counting roots of f or f'.** Between any two roots of f sits a root of f'.
- **Weighted-coefficient hypotheses.** If $\sum \frac{a_k}{k+1} = 0$, then $\sum a_k x^k$ has a root in $(0,1)$, because the antiderivative $F(x) = \sum \frac{a_k}{k+1} x^{k+1}$ satisfies $F(0) = F(1) = 0$.
- **Conclusions of the form $f'(c) = \lambda f(c)$.** Use the auxiliary function $g(x) = e^{-\lambda x} f(x)$; then $g' = e^{-\lambda x}(f' - \lambda f)$, and Rolle on g gives the result.
- **Iterated Rolle.** If f has $n+1$ equal values, $f^{(n)}$ has a zero. The basis of polynomial-interpolation error bounds.

Canonical applications.

- Show $x^5 + x^3 + x + 1 = 0$ has *exactly* one real root: IVT for existence, Rolle for uniqueness (since $f' > 0$ everywhere).
- Locate roots of derivatives of polynomials with simple factored form, like $f(x) = (x-1)(x-2)(x-3)(x-4)$ — root of f' in each gap.
- Gauss–Lucas theorem (in spirit): roots of f' lie in the convex hull of roots of f.

3 Lagrange's Mean Value Theorem (MVT)

Theorem (Lagrange)**.** *If f is continuous on $[a,b]$ and differentiable on (a,b), then*

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \text{for some } c \in (a,b).$$

Intuition. The instantaneous rate equals the average rate, somewhere. A car averaging 60 km/h must, at some instant, read 60 on the speedometer. Geometrically: tilt the chord horizontal and Rolle reappears — this is how the theorem is proved.

When to reach for it. Whenever you must compare f at two points using only information about f' in between. Typical signals:

- Inequalities involving ln, e^x, arctan, sin — "bound the difference" problems.
- Bounding $|f(b) - f(a)|$ given a bound $|f'| \le M$ (Lipschitz estimates).
- Proving f is monotonic, constant, or Lipschitz from properties of f'.
- Showing that a one-sided limit of f' at a point equals the one-sided derivative there.

Canonical applications.

- **Logarithmic bounds:** $\frac{b-a}{b} < \ln \frac{b}{a} < \frac{b-a}{a}$ for $0 < a < b$, from MVT applied to ln.
- **Lipschitz from bounded derivative:** $|f'| \le M \Rightarrow |f(x) - f(y)| \le M|x - y|$.
- **Sine inequality:** $|\sin x - \sin y| \le |x - y|$ in one line.
- **Tangent-line bounds:** $e^x \ge 1 + x$, with equality only at $x = 0$ — the prototypical convexity/Taylor estimate.
- **Derivative-limit theorem:** If $\lim_{x \to a^+} f'(x) = L$ exists, then $f'(a^+) = L$.

4 Cauchy's Mean Value Theorem

Theorem (Cauchy)**.** *If f, g are continuous on $[a,b]$ and differentiable on (a,b) with $g' \ne 0$, then*

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)} \quad \text{for some } c \in (a,b).$$

Intuition. Two runners' position functions $f(t)$ and $g(t)$: at some moment, the ratio of their instantaneous speeds equals the ratio of their total displacements. Setting $g(x) = x$ recovers Lagrange, so Cauchy is strictly more general.

When to reach for it. Whenever the quantity of interest is a *ratio*, or the situation involves comparing the growth of two functions. Signals:

- Expressions of the form $\frac{f(b) - f(a)}{g(b) - g(a)}$.
- $\frac{0}{0}$ indeterminate forms (L'Hôpital is Cauchy with a limit).
- Conclusions involving products $f'(c)g(c)$ or $f(c)g'(c)$.

Canonical applications.

- **L'Hôpital's rule:** The textbook computational shortcut is literally Cauchy's MVT plus a limit.
- **Comparison of two functions agreeing at endpoints:** If $f(0) = g(0) = 0$ and $f(1) = g(1) = 1$, then $f'(c) = g'(c)$ for some $c \in (0, 1)$.
- **Asymmetric expressions like** $\dfrac{af(b) - bf(a)}{a - b} = f(c) - cf'(c)$**:** solved by choosing the right pair, here $F(x) = \dfrac{f(x)}{x}$ and $G(x) = \dfrac{1}{x}$.
- **Generalised mean value identities:** the product form $[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c)$ avoids the $g' \neq 0$ hypothesis.

5 Quick-reference cheat sheets

Which theorem, by signal

Signal	Theorem
Prove a root, fixed point, or meeting exists	**IVT**
Show two functions agree somewhere	**IVT** on $h = f - g$
Discontinuous function has IVP	**Darboux** (if it's a derivative)
Count roots of f or f'; $f(a) = f(b)$	**Rolle**
Weighted sum of coefficients equals zero	**Rolle** on the antiderivative
$f^{(n)}(c) = 0$ from several vanishing points	Repeated **Rolle**
Bound $f(b) - f(a)$ from bounds on f'	**Lagrange**
Prove f is Lipschitz, monotonic, or constant	**Lagrange**
Inequalities with ln, e^x, arctan, sin	**Lagrange**
f' has a limit at a point $\Rightarrow$ derivative there	**Lagrange** (+ squeeze)
Compare growth of two functions (a ratio)	**Cauchy**
$\frac{0}{0}$ indeterminate; L'Hôpital flavour	**Cauchy**
Two functions agree at endpoints, compare f', g'	**Cauchy**

Auxiliary-function transformations

Conclusion sought	Auxiliary function	Why it works
$f'(c) = 0$	$g(x) = f(x)$	Rolle directly.
$f'(c) = \lambda$	$g(x) = f(x) - \lambda x$	Chord becomes horizontal.
$f'(c) = \frac{f(b)-f(a)}{b-a}$	$g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$	Proves the MVT itself.
$f'(c) = f(c)$	$g(x) = e^{-x}f(x)$	$g' = e^{-x}(f' - f)$.
$f'(c) = \lambda f(c)$	$g(x) = e^{-\lambda x}f(x)$	$g' = e^{-\lambda x}(f' - \lambda f)$.
$f'(c) + p(c)f(c) = 0$	$g(x) = e^{\int p} \cdot f(x)$	Integrating factor (ODE).
$cf'(c) = f(c)$	$g(x) = \frac{f(x)}{x}$	$g' = \frac{xf'-f}{x^2}$.
$cf'(c) = kf(c)$	$g(x) = \frac{f(x)}{x^k}$	$g' = \frac{xf'-kf}{x^{k+1}}$.
$f'(c)g(c) = f(c)g'(c)$	$h(x) = \frac{f(x)}{g(x)}$	Quotient rule, or apply Cauchy.

The single most useful move: when the conclusion looks like "$f'(c) = \lambda f(c)$" or any rearrangement, multiply by $e^{-\lambda x}$ and apply Rolle.

The big picture

- **IVT** — existence under continuity. Build $h = f - g$ and seek a sign change.
- **Rolle** — the parent of all MVT. Every Rolle problem is a hunt for an interior extremum, often via a clever auxiliary function.
- **Lagrange** — converts f'-bounds into f-bounds. The workhorse for inequalities and Lipschitz estimates.
- **Cauchy** — compares two functions' growth as a ratio. Reach for it whenever the problem is intrinsically about a ratio.
- **Darboux** — the surprise: derivatives cannot skip values, even when discontinuous.

All four MVTs are Rolle in different costumes. Lagrange tilts the chord, Cauchy parametrises, Darboux extends the interior-extremum idea to derivatives that needn't be continuous.